AWS VPC PEERING LAB

Configuration, Routing and Connectivity Validation

Evidence-based lab report | Screenshots captured 17–18 October 2026

1. Executive Summary

This lab configured VPC peering between two Amazon VPCs in AWS Mumbai (ap-south-1). VPC A (ewb-vpc-a) uses 10.10.0.0/16 and VPC B (ewb-vpc-b) uses 10.20.0.0/16. The peering connection reached Active and reciprocal peer-CIDR routes were configured. Security-group rules were set for ICMP and SSH testing.

Prior terminal testing recorded a successful ping to 10.20.1.172 (4 packets received, 0% loss) and a TCP/22 reachability probe. SSH authentication was not confirmed: the login attempt returned a public-key permission error. Network reachability is therefore verified, but successful SSH login remains unverified.

2. Objectives

• Create two VPCs with non-overlapping CIDR ranges.

• Create subnets and review route-table associations.

• Create and activate VPC peering.

• Configure reciprocal routes and security-group access.

• Validate private-IP reachability and document limitations.

3. Architecture and Address Plan

VPC A — ewb-vpc-a — 10.10.0.0/16: Source/test VPC

VPC B — ewb-vpc-b — 10.20.0.0/16: Peer/target VPC

VPC A public subnet — 10.10.1.0/24: Public test instance subnet

VPC A private subnet — 10.10.2.0/24: Private subnet

VPC B private subnet — 10.20.1.0/24: Target instance subnet

VPC B target instance — 10.20.1.172: Private-IP connectivity target

VPC peering connection — pcx-06399d954e78df776: Direct VPC-to-VPC route target

4. Methodology

Plan address ranges: Use 10.10.0.0/16 for VPC A and 10.20.0.0/16 for VPC B. The ranges do not overlap.

Create networking resources: Create VPCs and subnets, then inspect subnet-to-route-table associations.

Create peering: Request peering between VPC A and VPC B and accept it so the connection becomes active.

Add reciprocal routes: In the relevant route tables for VPC A, route 10.20.0.0/16 to the peering connection. In the relevant route table for VPC B, route 10.10.0.0/16 to the same connection.

Configure security groups: Permit only the needed test traffic from the peer CIDR, including ICMP and TCP/22 for the lab.

Test connectivity: Ping the VPC B private IP from VPC A. A successful ping validates ICMP reachability; a TCP/22 probe validates that the port is reachable but does not establish SSH authentication.

5. Results and Validation

VPC peering status: Active (based on lab context and console evidence).

Route configuration: Reciprocal routes were configured for the peer CIDR in the relevant route tables.

ICMP test: Successful: 4/4 replies from 10.20.1.172; 0% packet loss (reported during the lab).

TCP/22 reachability: Port reachability probe succeeded (reported during the lab).

SSH authentication: Not verified. Login attempt failed with public-key authentication error; key-pair/private-key access must be resolved before claiming SSH login success.

Cost-control state: The four EC2 instances were later reported as stopped. The October billing PDF showed an estimated total of USD 0.00 as of 18 October 2026; estimates can change and stopped instances may still have billable attached resources.

6. Troubleshooting and Safety Notes

• VPC peering requires routes on both sides; it does not automatically create them.

• Security groups and network ACLs must allow the intended traffic.

• TCP/22 reachability is not proof of successful SSH authentication.

• Keep private keys and account secrets out of public posts.

• Stopped instances may still have billable EBS volumes, snapshots, or allocated public IP resources.

7. Conclusion

The lab demonstrated the core VPC peering workflow: non-overlapping address spaces, an active peering connection, reciprocal routes, and security-group configuration. The reported ping passed; SSH authentication remains unverified.

8. Evidence Appendix

Figure 1. VPC peering connection overview

The peering connection was created between the two VPCs. AWS documentation notes that a peering connection must be active and route tables on both sides must direct peer-CIDR traffic to it.

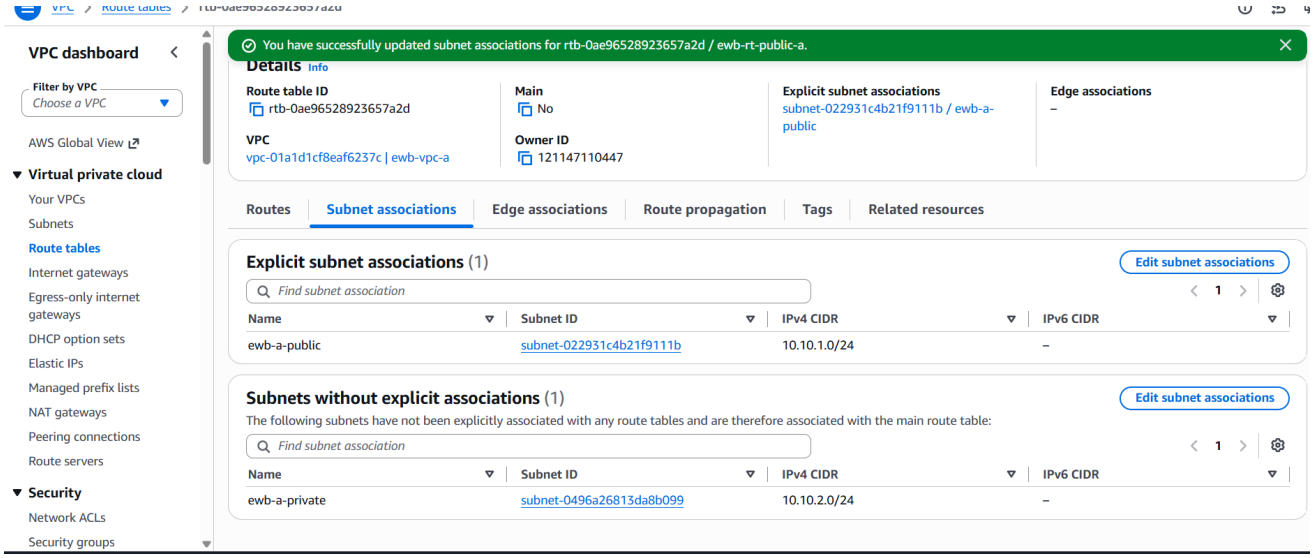


Figure 2. Route table configuration

Route table evidence showing the route-table workflow. The final setup used routes for the opposite VPC CIDR through the peering connection on both sides.

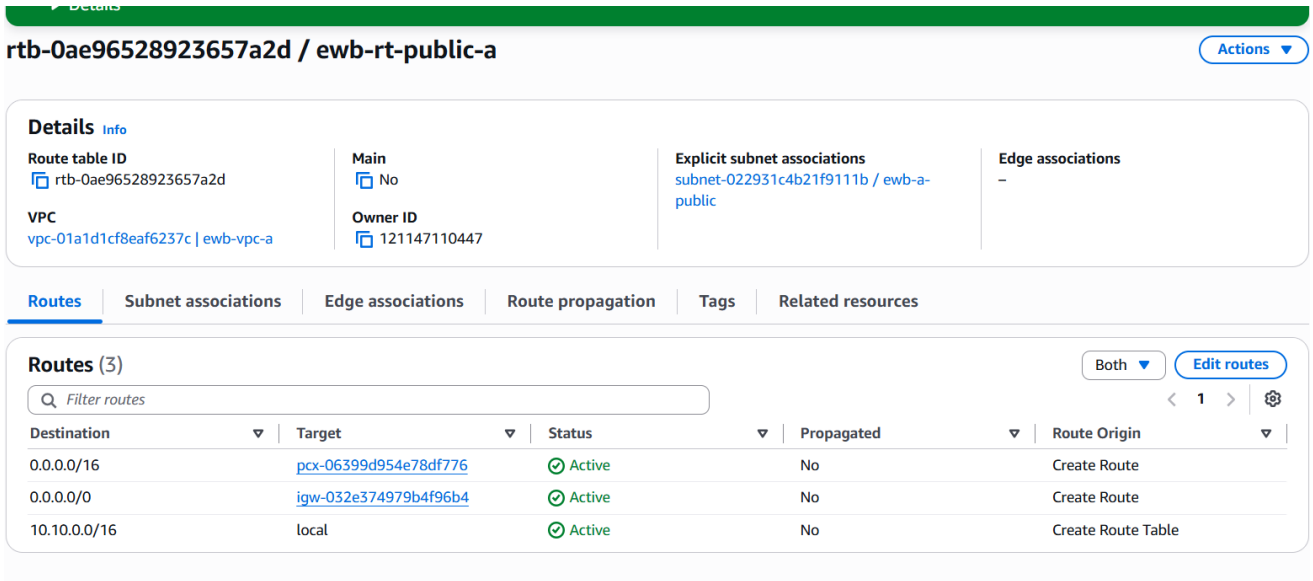


Figure 3. Subnet and route-table association

Subnet/route-table association evidence, relevant because each subnet uses its explicitly associated route table or the VPC main route table.

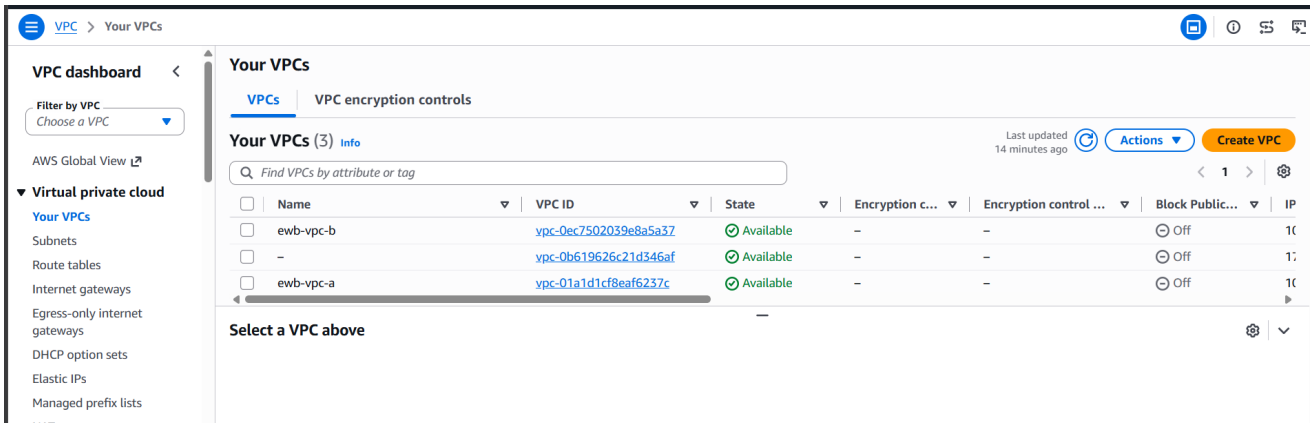


Figure 4. VPC and subnet configuration

VPC networking configuration evidence, supporting the documented use of non-overlapping VPC address ranges.

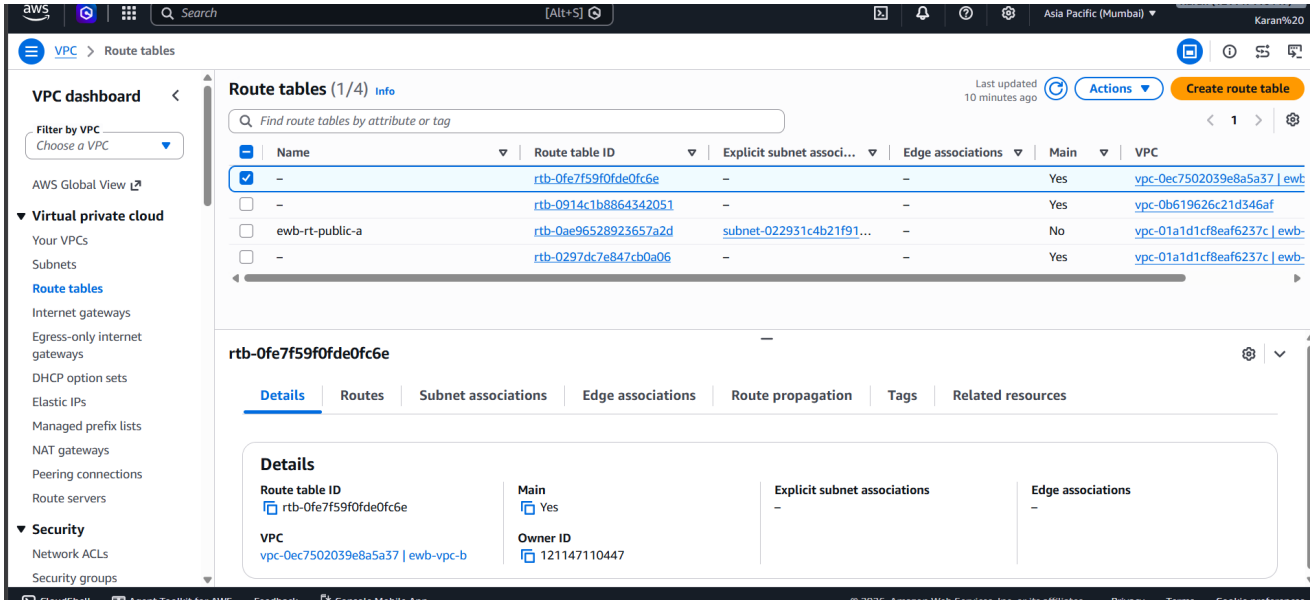


Figure 5. VPC resource overview

A console overview of VPC resources created during the lab.

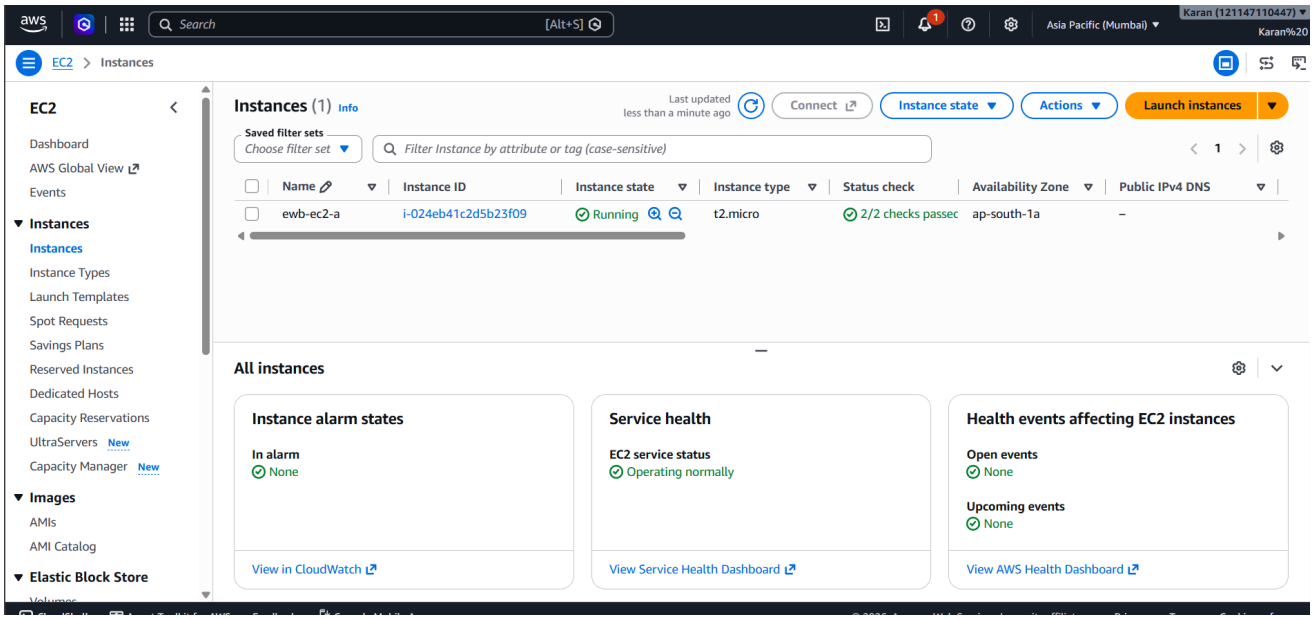


Figure 6. Network configuration workflow

Configuration evidence from the networking workflow; included as supporting context rather than a standalone connectivity test.

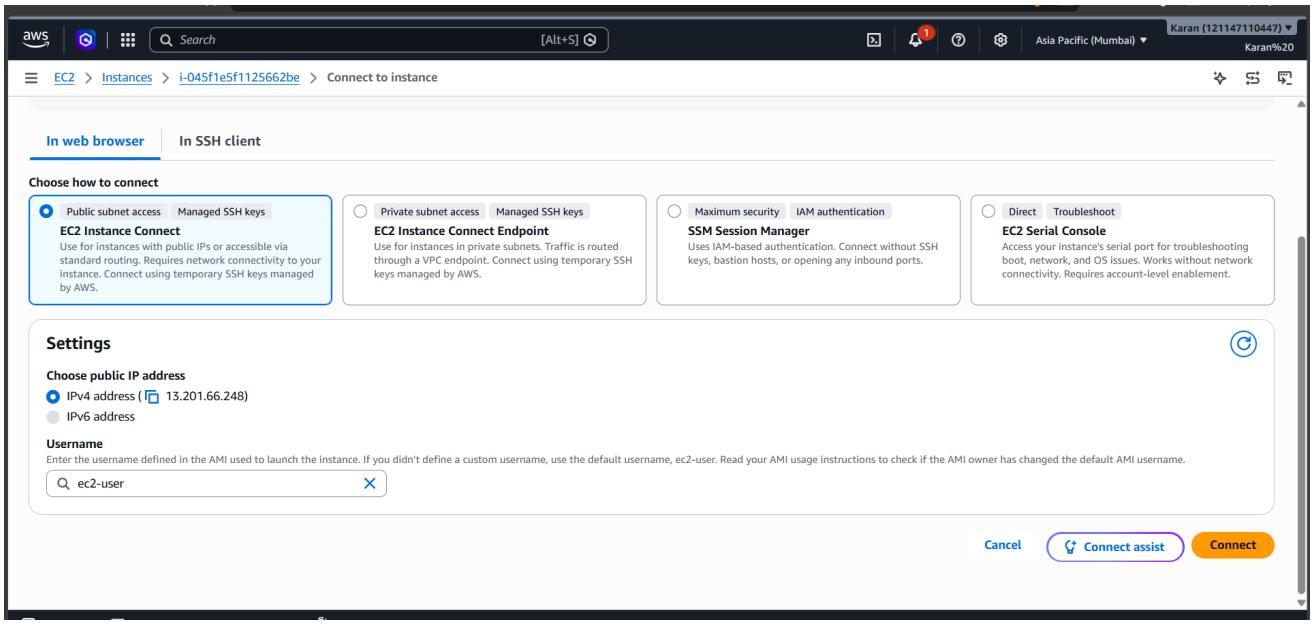


Figure 7. Security group configuration

Security group configuration evidence. The lab used inbound ICMP and SSH rules from the peer VPC CIDR for the test.

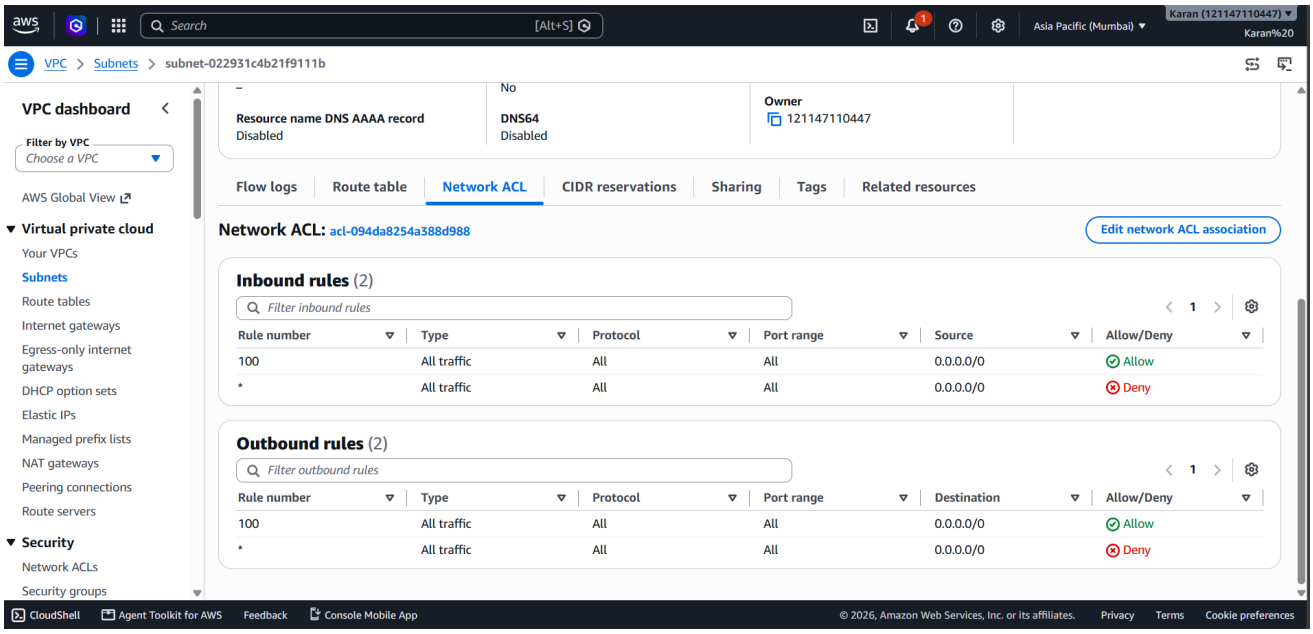


Figure 8. EC2 launch/configuration workflow

EC2 configuration evidence from the lab workflow.

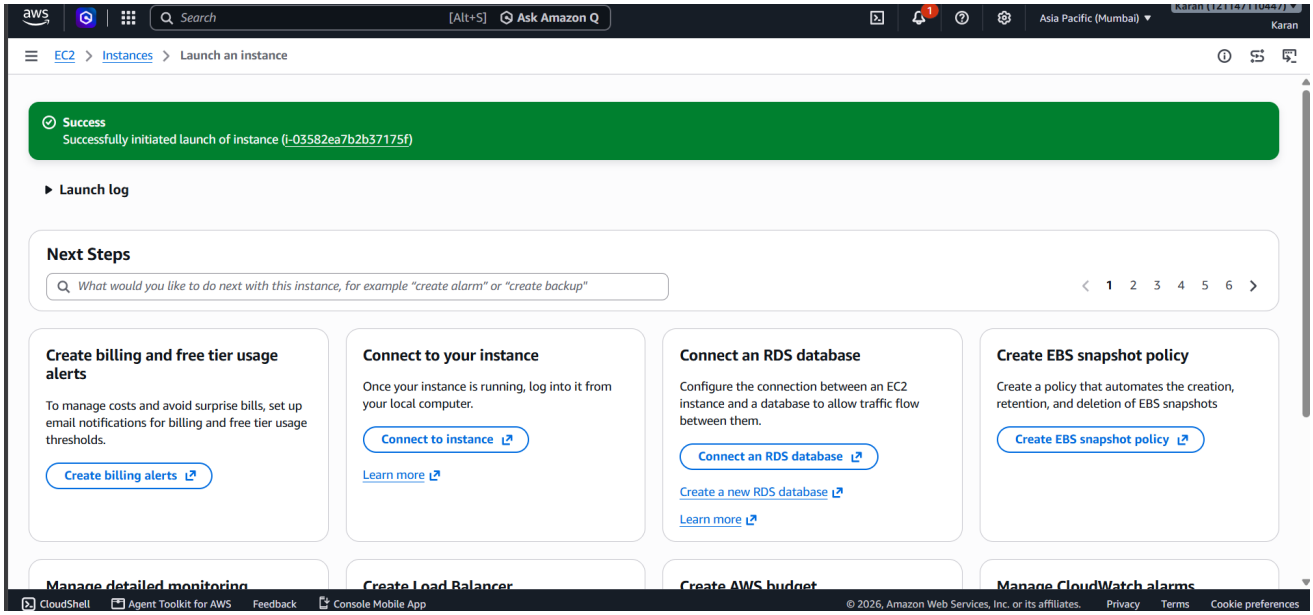


Figure 9. EC2/VPC resource view

Resource overview from the AWS console.

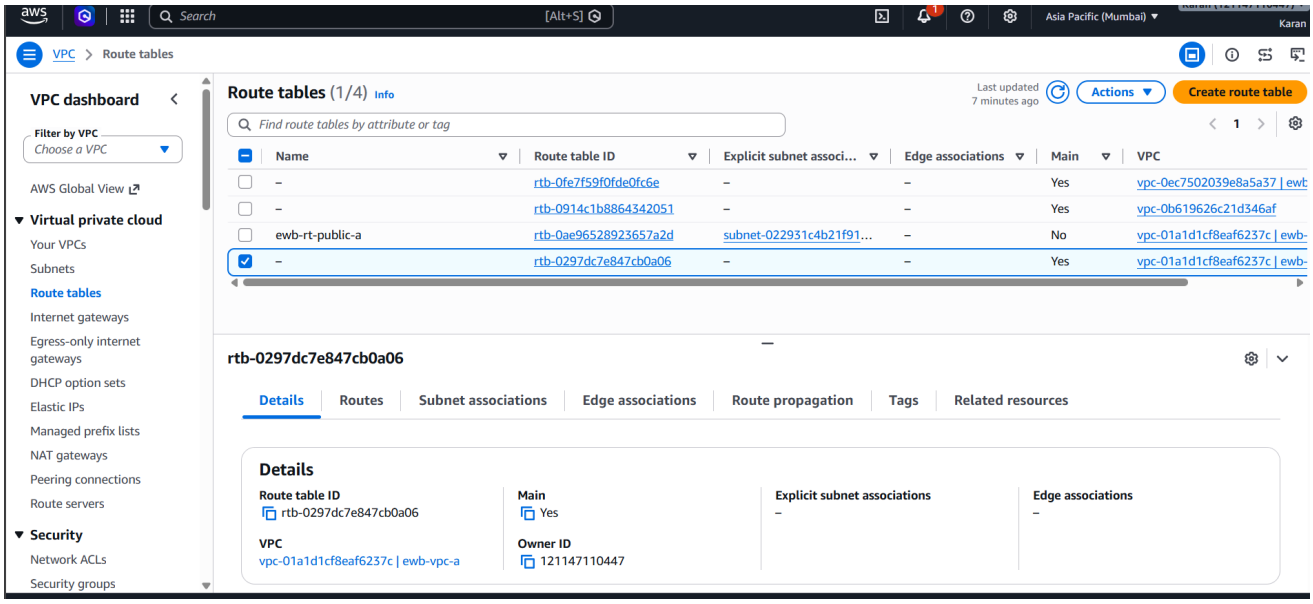


Figure 10. EC2 terminal session

Terminal access to an Amazon Linux 2023 instance. This screenshot by itself does not prove cross-VPC connectivity.



Figure 11. Route table review

Route-table review evidence captured after the initial setup.

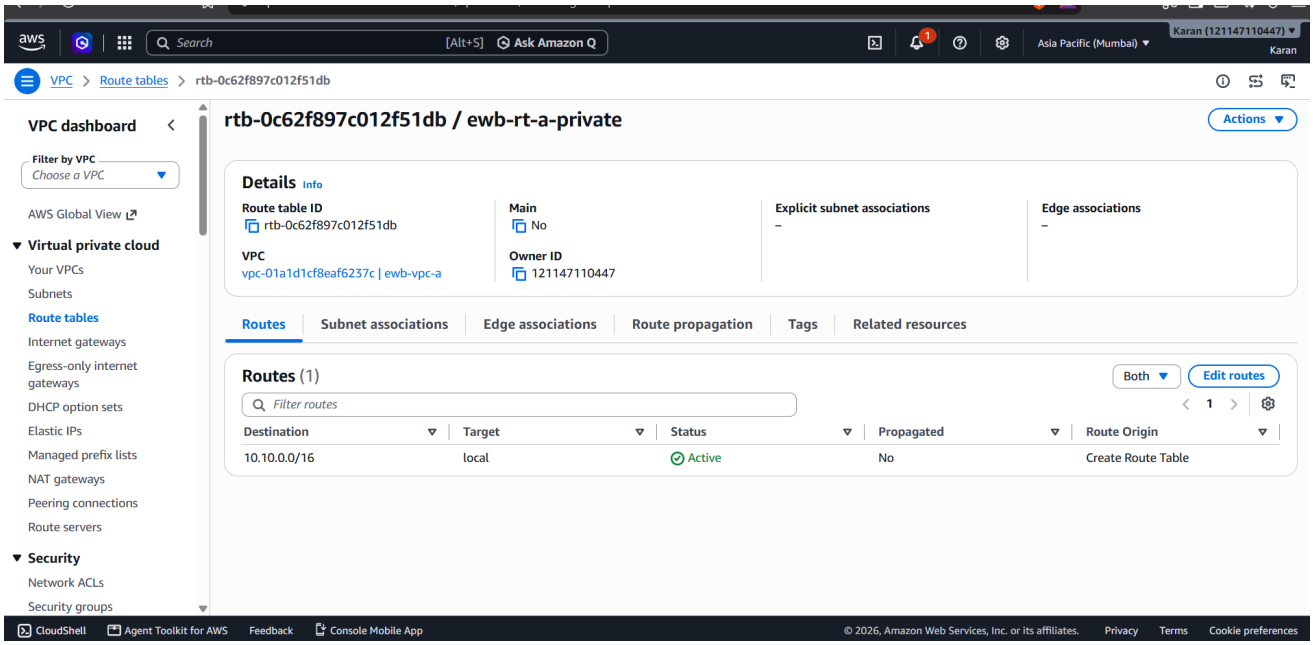
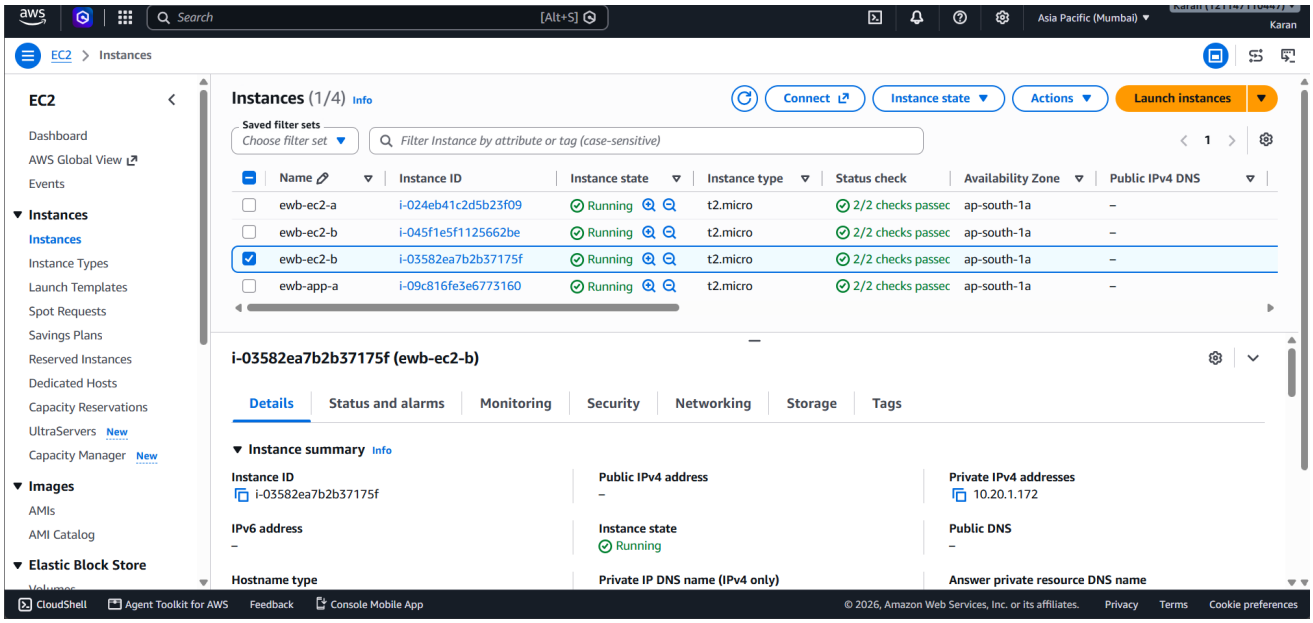


Figure 12. Final networking/resource review

Later review of the configured networking resources.



9. References

AWS: How VPC peering connections work

AWS: Update your route tables for a VPC peering connection